

Neural Activity Analysis of Athletes in Baseball Batting Using Virtual Reality and EEG

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Figure 1: A participant wearing an EEG cap and HMD performs the button-pressing task in a batting stance. The VR baseball stimuli are synchronized with EEG signal acquisition to ensure precise event-related data collection.

ABSTRACT

This study proposes a measurement system for neural processes in baseball batting using virtual reality (VR). Investigating neural processes of baseball batting in real-world environments is challenging due to physical constraints. To address this, we integrated a baseball VR environment with electroencephalography (EEG) recording to track brain activity in a setting that mimics actual game conditions. We conducted a pitch-type recognition task to compare neural responses between experienced and non-experienced groups. Our results revealed that experienced players exhibited significantly more pronounced positive amplitudes, demonstrating that expertise-related differences in perceptual-cognitive processes are quantifiable in immersive settings. These findings suggest that EEG metrics in VR environments can serve as effective objective indicators for evaluating a batter's expertise and perceptual-cognitive abilities.

Index Terms: Virtual Reality, Pitch-type Recognition, EEG, Sports Expertise

1 INTRODUCTION

Batting in baseball is one of the most demanding motor tasks in sports, requiring rapid visual processing and decision-making within 150–200 ms [1]. While the optimization of neural processes is essential for achieving proficiency, investigating these underlying mechanisms in real-world environments is challenging due to issues with experimental reproducibility and safety concerns.

To address these challenges, this study utilizes a virtual reality (VR) environment integrated with electroencephalography (EEG). By implementing a pitch-type recognition task and comparing EEG data between different skill levels, we investigated the neural characteristics associated with athletic expertise. Our findings suggest that VR provides a practical and effective platform for sports science, offering new possibilities for analyzing the neural correlates of athletic performance.

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2 RELATED WORKS

Previous research utilizing EEG has shown that expert players exhibit enhanced perception–action coupling and more efficient visual information processing compared to novices [5, 11] in pitch-type recognition, with stimulus–response compatibility playing a key role in mediating these elite skills [6]. Nevertheless, traditional studies often rely on simplified 2D monitors that lack ecological validity, as they cannot replicate the 3D spatial perception required in a real batting situation.

Virtual Reality (VR) has emerged as an effective tool to bridge this gap, offering a controlled yet immersive environment that replicates the spatial and visual demands of sport [2, 10]. VR systems replicating batting situations have proven to be effective for training, improving on-field performance in both baseball and softball by providing realistic pitching simulations [12, 7]. By providing a high level of ecological validity, VR allows for the study of complex athletic behaviors while maintaining the experimental control required for the collection of high-quality physiological data.

3 EXPERIMENTAL SETTINGS

3.1 System Overview

We implemented an experimental setup as shown in Figure 1. The VR baseball environment was developed using the Unity 3D engine (2018.4.14f). Visual stimuli were presented via a Valve Index Head-Mounted Display (HMD), operating at a consistent refresh rate of 90 Hz. The system continuously captured the HMD's spatial position and orientation, ensuring the participant's real-time movement was accurately reflected within the virtual environment. To capture neural activity, we used an ANT Neuro eegoTM sports EEG system equipped with 63 active Ag/AgCl electrodes. EEG signals were recorded at a sampling rate of 2048 Hz. The HMD was carefully fitted over the EEG cap, ensuring that electrode impedance levels were maintained below 20 kΩ throughout the duration of the experiment. To ensure precise temporal alignment between the VR environment and the neural data, we utilized a CONTEC USB-based data I/O unit to record VR events timing as digital event markers in the EEG recording.

3.2 Participants

Twenty-four healthy participants (24 males; *Mean* = 22.3 ± 2.56 years) participated in the experiment. They were divided

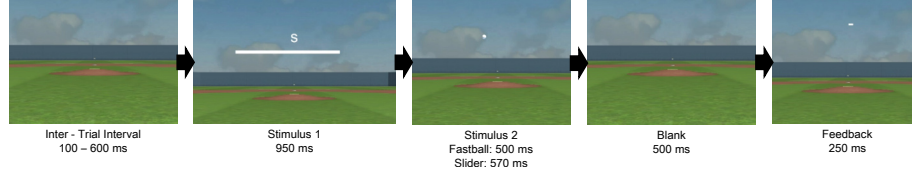


Figure 2: Trial sequence in the virtual baseball task.

into two groups: a baseball-experienced group ($n = 12$, $M = 20.9 \pm 1.44$ years) and a non-experienced group ($n = 12$, $M = 23.8 \pm 2.65$ years). The experienced group consisted of individuals who had played baseball in organized teams, such as junior high and high school clubs, for at least six years. The non-experienced group consisted of individuals with no formal baseball experience. All participants reported normal or corrected-to-normal vision and provided oral informed consent prior to their participation. The study was approved by the Research Ethics Committee of the Faculty of Science and Technology, Keio University.

3.3 Task and Presented Stimuli

Participants performed a Go/No-Go button-pressing task while standing in the batter's box within the VR environment, maintaining the posture shown in Figure 1. The experimental paradigm, adapted from [5], is illustrated in Figure 2. Each trial began with a pitch-type cue (S1), which was displayed directly above the pitcher's mound within the VR environment. This cue was accompanied by a timing bar that shrunk over a period of 950 ms, providing a visual countdown until the pitch delivery. Immediately following S1, the ball trajectory (S2) was presented as either a fastball (500 ms travel time) or a slider (570 ms). Participants were instructed to press a button as quickly as possible if the S2 trajectory matched the S1 cue (Go condition) and to withhold their response if they did not match (No-Go condition). Visual feedback ("+" for correct; "-" for incorrect) was provided after each response, with trial distribution ratios detailed in Table 1.

Table 1: Experimental conditions for each participant

		S2	
		Fastball (F)	Slider (S)
S1	Fastball (F)	FF condition (Go) 135 trials	FS condition (No-Go) 90 trials
	Slider (S)	SF condition (No-Go) 90 trials	SS condition (Go) 135 trials

3.4 EEG Data Processing and Statistics

EEG data pre-processing was conducted using the FieldTrip toolbox [8] in MATLAB (R2024b). The continuous EEG data were first band-pass filtered between 1 and 30 Hz using a zero-phase Finite Impulse Response filter. Data were then segmented into epochs from -1500 to 2000 ms with the onset of S2 defined as 0 ms. Each epoch was baseline-corrected from -200 to 0 ms. Following the rejection of noisy trials via visual inspection, an Independent Component Analysis (ICA) [3] was performed to remove physiological artifacts. Independent components (ICs) [9] were automatically classified using the ICLabel plugin. Artifact rejection was performed based on class-specific probability thresholds: ICs were removed if the probability of being categorized as "Eye," "Heart," "Line Noise," or "Channel Noise" was $\geq 80\%$, or if the probability of the "Muscle" category was $\geq 30\%$.

Following pre-processing, only trials with correct responses were included in the analysis. Time-series amplitude data were concatenated across trials for each condition. To identify significant

temporal windows of difference between groups, cluster-based permutation test (CBPT) [4] was performed for each electrode using 10000 permutations.

4 RESULT AND DISCUSSION

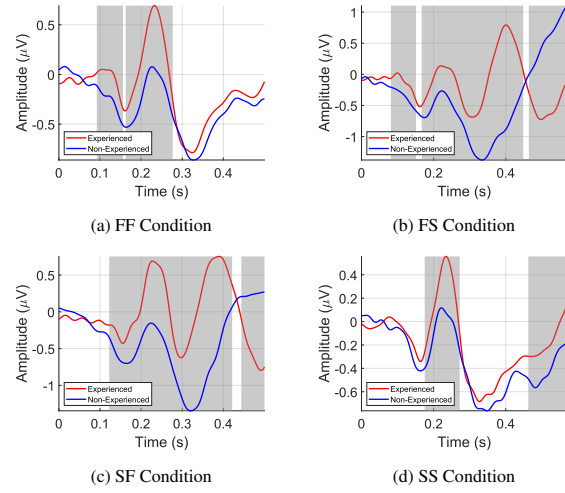


Figure 3: Grand-averaged amplitude waveforms and significant clusters at Cz electrode under each condition (FF, FS, SF, SS). The waveforms represent the averaged neural responses for experienced (red line) and non-experienced (blue line) groups during the S2 presentation. Shaded gray boxes indicate temporal cluster regions where a significant difference in amplitude between the two groups was identified using CBPT.

The results of the grand-averaged amplitude waveforms at the Cz electrode are presented in Figure 3. Across all conditions, the experienced group exhibited a significantly more positive deflection compared to the non-experienced group in the temporal window around 200–250 ms. In addition, in the No-Go conditions (FS and SF), the experienced group showed a significantly higher positive amplitude at approximately 400 ms. These findings suggest distinct neural processing between the two groups during pitch-type recognition and the mechanisms of motor inhibition.

5 CONCLUSION

In this study, we proposed a pitch-type recognition task within a VR baseball environment and analyzed EEG data to identify distinct differences in neural processing between experienced and non-experienced players. The results revealed that experienced players exhibit significantly more pronounced positive amplitude deflections during the visual processing of pitch trajectories and motor control. These findings demonstrate that EEG metrics in VR environments can serve as effective objective indicators for evaluating a batter's expertise and perceptual-cognitive abilities.

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